


# FACTORICS PROJECT PROPOSAL

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GITHUB LINK: HTTPS://GITHUB.COM/NHAHUB/NHA-5-316

# OVERVIEW

This year, A new Egyptian government-backed fund was launched to support defaulted and about-to-default factories in Egypt. That's how the idea of creating of FactoRics was inspired.

FactoRics is an industrial downtime & efficiency intelligence platform. The platform tries to predict the minor and major dropping in production efficiency.

The integrated industrial data analysis designed to process production sheets. The Data contain four sheets (Rows x Columns):

- ProcessedHourlyData (272 x 4)
- DailyOperationSummary (59 x 11)
- HourlyOperationBreakdown (265 x 7)
- DowntimeEventLog (1388 x 5)

The system relies on an End-to-End Data Pipeline starting with data storage and querying via SQL, advanced cleaning, feature engineering, and AI modeling via Python, followed by a Dual-BI Layer combining Power BI for daily operational monitoring and Tableau for strategic executive storytelling.

## THE OBJECTIVE

- **#1:** Monitoring hourly operations data in Egyptian factories.
- **#2:** Analyzing the dataset statistically to determine the dangerous efficiency levels daily.
- **#3:** Predicting efficiency and losses in the future.

## TOOLS & DESIGN

A structured End-to-End Data Pipeline consisting of 4 integrated layers:

- **#1:** Data Ingestion & Storage Layer (SQL)
- **#2:** Data Processing & Machine Learning Layer (Python)
- **#3:** Operational Monitoring Layer (Power BI)
- **#4:** Strategic & Financial Analytics Layer (Tableau)

## TECHNOLOGIES TO USE

- **SQL:** Raw data storage, required querying.
- **Python (ex: Pandas, NumPy, Scikit-Learn):** Procedural cleaning, missing value handling (NaN/NaT), outlier management (IQR Winsorization), ML modeling.

- **Power BI:** Operational Dashboards for tracking micro-stoppages and daily downtime for maintenance and production engineers.
- **Tableau:** Strategic & Financial Data Storytelling dashboards for executive leadership to monitor overall financial losses and long-term trends.

## RESPONSIBILITIES

This project is a solo project under DEPI initiative by Aisha Mohamed Kamel. Fully responsible for executing all project phases independently.

1. SQL database management
2. Python scripts and ML models
3. Power BI and Tableau dashboards
4. Documentation
5. GitHub deployment

## FURTHER ANALYSIS

By connecting more economic and financial data, the platform can be used to determine the best financial solutions for the factories facing bankruptcy. This approach may not be expanded due to the insufficient time sources.

FactoRics

# PROJECT TIMELINE

I expect this project to follow this timeline:

## WEEK 1: SQL PART

- **Step #1:** Setting up the SQL database.
- **Step #2:** Storing Row Data
- **Step #3:** Creating the related queries and views, if needed.
- **Deliverables: SQL query file.**

## WEEK 2: PYTHON - PART 1 - (EXPLORATION AND CLEANING)

- **Step #1:** Building Initial Python script for data and column standardization.
- **Step #2:** Handling datetime errors and missing values (NaN/NaT).
- **Step #3:** Managing outliers via the suitable techniques.
- **Step #4:** Determination of Data Analysis Questions.
- **Deliverables: Python cleaning code, set of analysis questions that can be answered via the dataset analysis in the following week (Week 3).**

## WEEK 3: PYTHON - PART 2 - (ANALYTICS AND ML)

- **Step #1:** Trainning model for efficiency prediction.
- **Step #2:** Testing the codes.
- **Step #3:** Exploring the difference of the past and future (predicted) data.
- **Deliverables: full python script, visual plots answering analytical questions, and visual plots exploring how accurate the ML model.**

## WEEK 4: DASHBOARDING AND DOCUMENTATION

- **Step #1:** Designing the operational dashboard on Power BI.
- **Step #2:** Designing the strategic dashboard on Tableau.
- **Step #3:** Finishing the documentation.
- **Deliverables: Power BI Dashboard, Tableau Dashboard, GitHub Technical Documentation, Project Presentation.**

FactoRics